Pimpri-Chinchwad Education Trust's
Pimpri-Chinchwad College of Engineering
(An Autonomous Institute)
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Assignment No 7

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Subject: Design and Analysis of Algorithms
Batch: A3
GitHub Link: https://github.com/SOHAM0007-CODER/DAA_ASSIGNMNET

1. Topic

Create a conflict-free university exam timetable using graph coloring. Each course is a vertex. Add an edge when a student takes both courses, then assign different exam slots to adjacent courses. Reduce slots where possible and study classroom-capacity constraints.

2. Course Objective

- Understand the computational complexity of algorithms.
- Select a suitable algorithm design strategy for a practical problem.

3. Course Outcome

- Model common-student conflicts and produce a valid Greedy coloring.
- Compare coloring strategies and explain how rooms affect a feasible timetable.

Theory

Graph coloring assigns a color to each vertex so that connected vertices have different colors. Here, colors represent time slots. The simple program reads course names and conflict pairs directly. It schedules courses in ID order using the first slot not already used by a scheduled conflicting course.

Color names represent time slots: Red, Blue, Green, etc. Colors are labels, not literal display ink. Twelve named colors are available; extra slots receive unique Color_13, Color_14, etc. Labels never wrap around to a conflicting existing color.

Working of the Algorithm

- 1. Read course names and all shared-student conflict pairs.
- 2. Mark colors already used by previously scheduled neighbors.
- 3. Assign the first legal color and print course-color assignments.
- 4. Group exams by color and check every conflict pair for distinct colors.

Complexity Analysis

Resetting the used-slot array for every course takes O(V squared) work; neighbor scans add O(E). Total time is O(V squared + E).

Time Complexity

Best Case: O(V squared + E).

Average Case: O(V squared + E).

Worst Case: O(V squared + E); course order can affect slot count.

Space Complexity

O(V + E) for adjacency lists, assigned slots and the temporary used-slot array.

Algorithm - Steps

- Step 1: Read course names and every conflicting course pair.
- Step 2: Process courses in order and assign the first legal slot.
- Step 3: Print the timetable; check that every conflict pair has different slots.

Pseudocode

```
set slot[course] = 0 for every course
for course u in ID order:
    mark every slot as unused
    for each neighbor v of u:
        if slot[v] is assigned: mark it used
    choose the smallest unused slot s
    slot[u] = s
print courses, assigned slots and total slots
```

Flow of Execution

START -> Read courses and conflicts -> Mark blocked slots -> Assign first legal slot -> Print timetable -> END.

6. Example

Courses: DAA, DBMS, OS, CN, AI, Maths. The saved conflicts produce Red: DAA, CN, Maths; Blue: DBMS, AI; Green: OS. Every conflict check passes. The DAA-DBMS-OS triangle proves at least three colors are needed in this example.

Welsh-Powell colors in decreasing degree order; DSATUR chooses the course with the most distinct neighbor colors. Both are heuristics. A room extension must ensure sufficient seats and prevent double-booking; extra slots may be needed. The simple code handles conflict coloring.

Conclusion

Greedy creates a valid course-color schedule and validates its conflict pairs. It does not generally guarantee the minimum colors. DSATUR, Welsh-Powell comparisons and classroom allocation remain extensions.